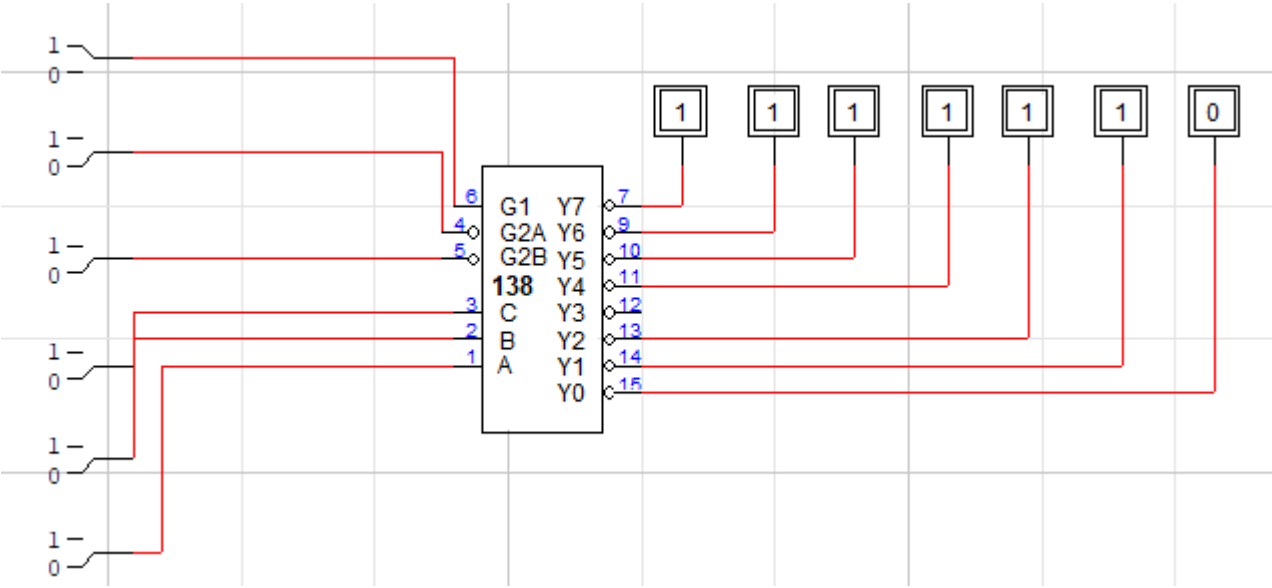
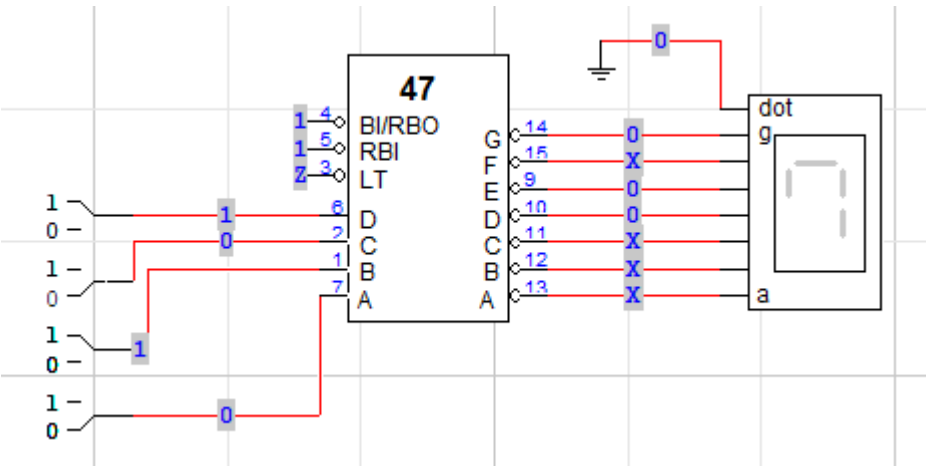


Task 1:



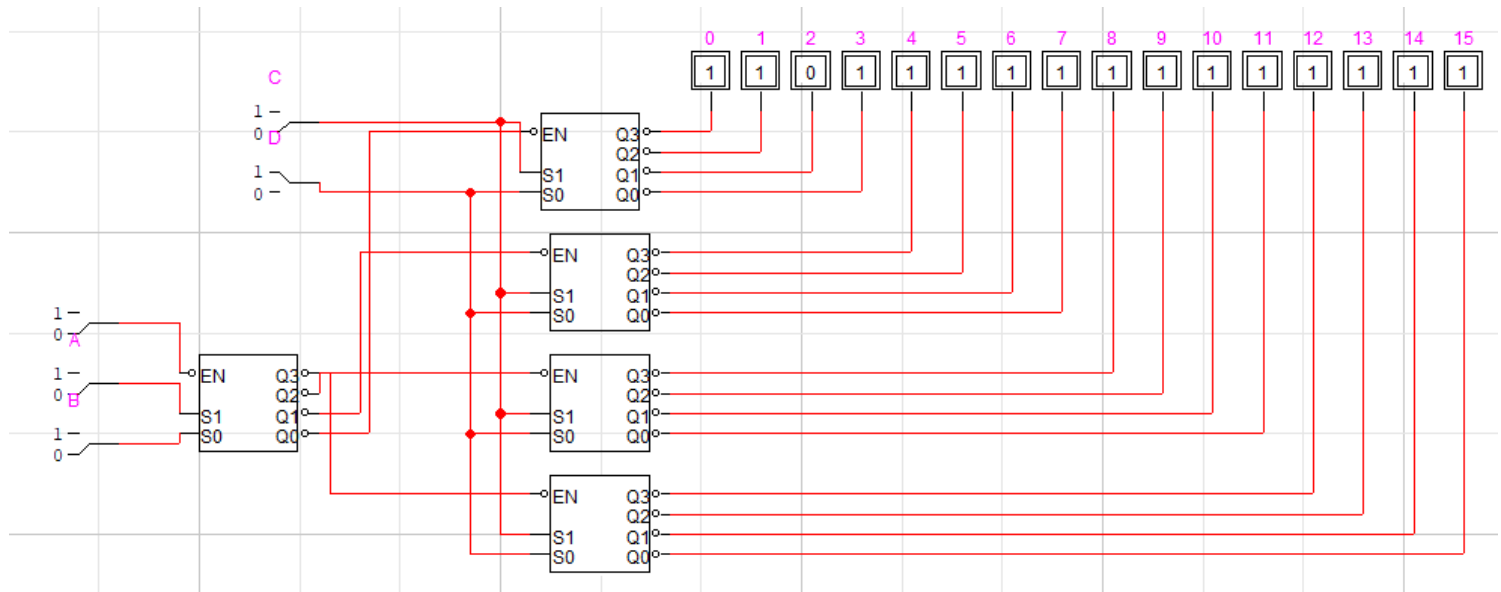
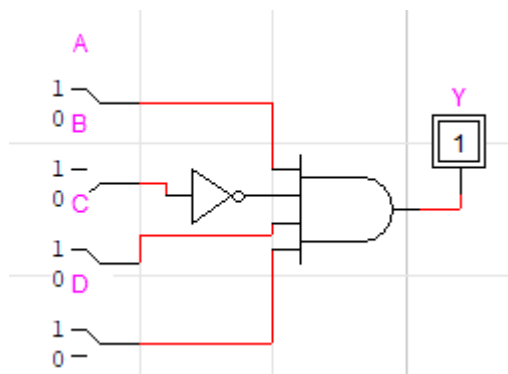
INPUTS						OUTPUTS							
Enable			Select										
G1	G2A'	G2B'	A2	A1	A0	Y0'	Y1'	Y2'	Y3'	Y4'	Y5'	Y6'	Y7'
1	0	0	0	0	0	0	1	1	1	1	1	1	1
1	0	0	0	0	1	1	0	1	1	1	1	1	1
1	0	0	0	1	0	1	1	0	1	1	1	1	1
1	0	0	0	1	1	1	1	1	0	1	1	1	1
1	0	0	1	0	0	1	1	1	1	0	1	1	1
1	0	0	1	0	1	1	1	1	1	1	0	1	1
1	0	0	1	1	0	1	1	1	1	1	1	0	1
1	0	0	1	1	1	1	1	1	1	1	1	1	0

Task 2:



INPUTS				OUTPUTS						
A3	A2	A1	A0	a'	b'	c'	d'	e'	f'	g'
0	0	0	0	0	0	0	0	0	0	1
0	0	0	1	1	0	0	1	1	1	1
0	0	1	0	0	0	1	0	0	1	0
0	0	1	1	0	0	0	0	1	1	0

0	1	0	0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	0	1	0	0
0	1	1	0	0	1	0	0	0	0	0
0	1	1	1	0	0	0	1	1	1	1
1	0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	1	0	0

Task 3:**Task 4:****Task 5:**

- a) 1110
- b) 1100
- c) 1111
- d) 1001

Task 6:

1st figure: Correct output should be 9. Circuit is working correctly.

2nd figure: Correct output should be 1. Display shows 0.

Possible faults:

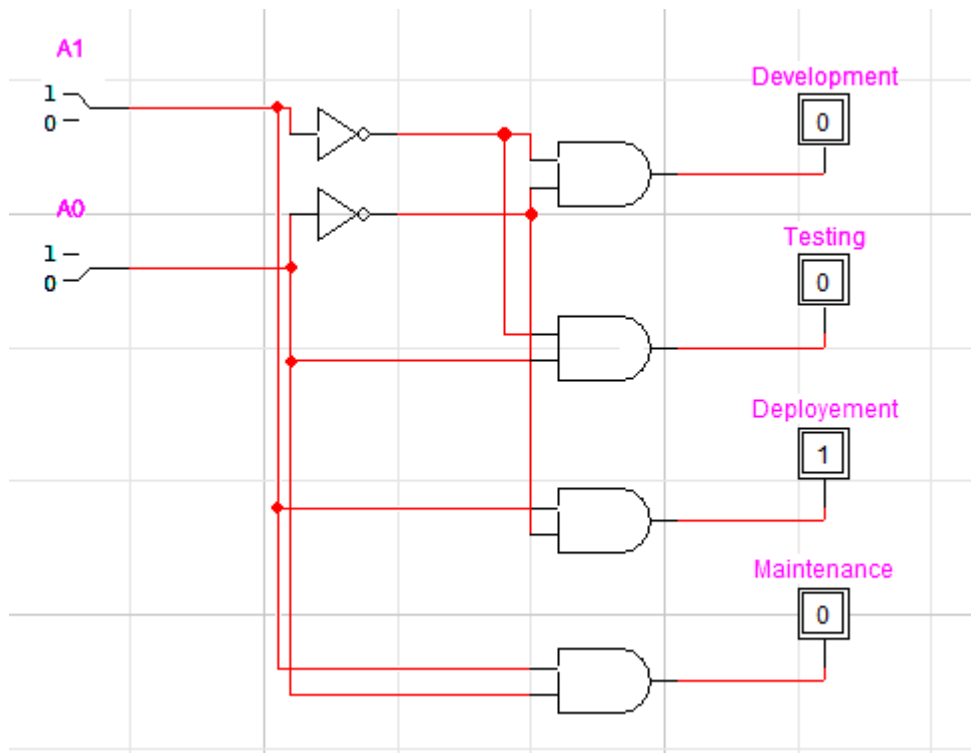
- Segments **a, d, e, f** stuck ON
- Decoder outputs for these segments shorted to LOW
- Wrong wiring between decoder and display

3rd figure: Correct output should be 6. Display shows 8.

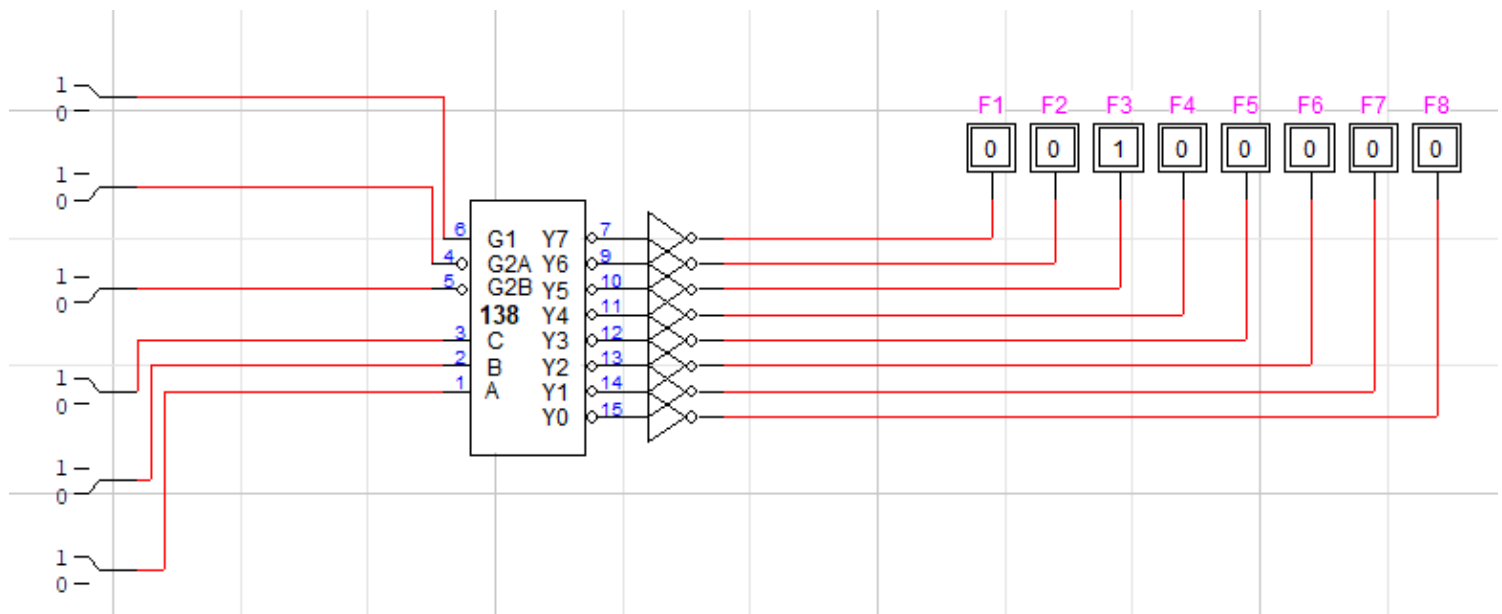
Possible faults:

- All outputs stuck LOW
- Faulty decoder IC
- Short circuit in outputs

Task 7:

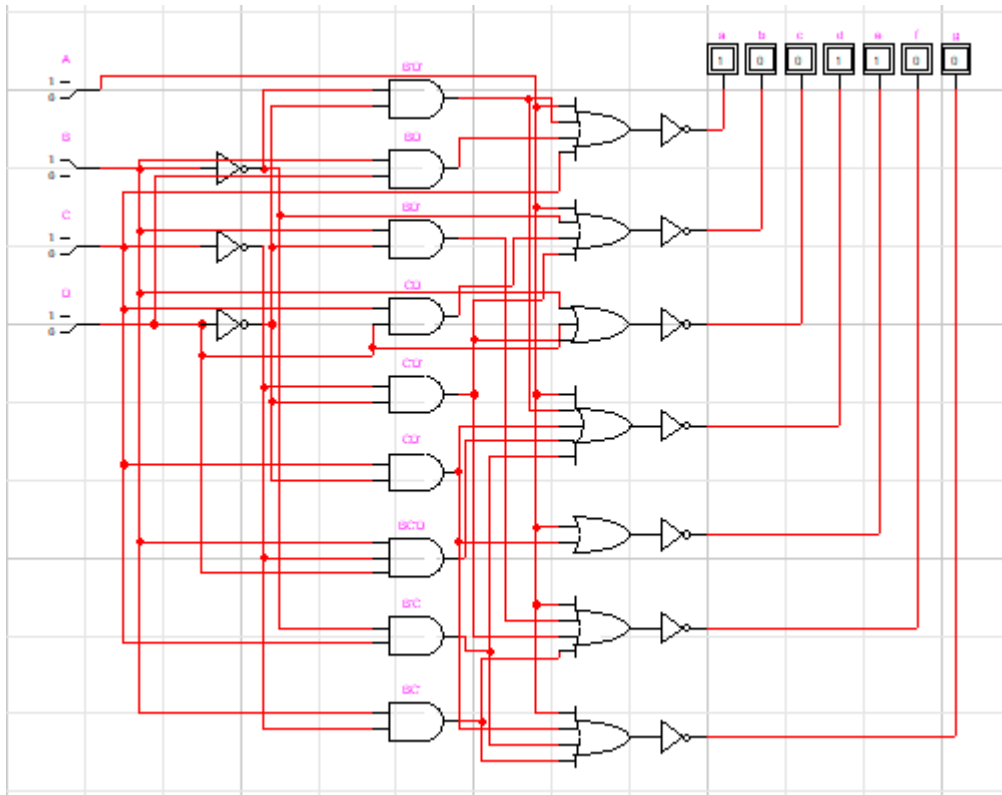


Task 8:



The outputs of 74LS138 are **active LOW**, meaning:

- Selected feature line = **0 (LOW)**
- But usually systems expect **1 (HIGH)** to enable.

Task 9:**Task 10:**